LESSON PLAN

GAS TURBINE ENGINE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various methods of thrust reversal and principles of operation.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various methods of thrust reversal and principles of operation.
5. Specific Objectives	13.1.5.1 Trainee will be able to Explain methods and principles of operation of thrust reversal.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of thrust reversal diagrams in IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of thrust reversal?
(b) Name one type of jet engine that utilizes thrust reversal.
(c) Why is thrust reversal critical during landing operations?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Cold Reverser Principle • Hot Reverser Principle • Thrust Reverser Components • Operational Sequence • Safety Limitations	L & Ds	PPT, IB	15 min
(b)	• Reverser Actuation Systems • Hydraulic vs Pneumatic • Locking Mechanisms • Exhaust Door Operation • Vectoring Nozzle Control	L & Ds	PPT, PDF	20 min
(c)	• Reverse Thrust Phases • Ground Idle Transition • Full Reverse Position • Retraction Criteria • System Failures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between cold and hot reversers.
(b) Describe the role of the locking mechanism in a thrust reverser.
(c) Under what conditions must thrust reversers be retracted?

12. Home Assignment and Follow Up	Prepare a brief note comparing the operational principles of thrust reversal in LP 02 (Lubrication system) context versus engine thrust management, focusing on how system reliability impacts engine performance.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Lubrication system main components and its purposes, operation of each components of system, Air cooling and sealing of engine main bearings.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min